

E14A.2(a) Calculate the resultant of two dipole moments of magnitude 1.5 D and 0.80 D that make an angle of 109.5° to each other.

E14A.4(a) What strength of electric field is required to induce an electric dipole moment of magnitude $1.0 \mu\text{D}$ in a molecule of polarizability volume $2.6 \times 10^{-30} \text{ m}^3$ (like CO_2)?

E14B.1(a) Calculate the molar energy required to reverse the direction of an H_2O molecule located 100 pm from a Li^+ ion. Take the magnitude of the dipole moment of water as 1.85 D.

E14B.2(a) Use eqn 14B.3b to calculate the molar potential energy of the dipolar interaction between two amide groups separated by 3.0 nm with $\theta = 45^\circ$ in a vacuum. Take $\mu_1 = \mu_2 = 2.7 \text{ D}$.
$$V = \frac{\mu_1 \mu_2 f(\theta)}{4\pi\epsilon_0 r^3} \quad f(\theta) = 1 - 3\cos^2\theta$$

Point dipole–point dipole interaction
[as in (4)] (14B.3b)

E14B.4(a) Calculate the average interaction energy for pairs of molecules in the gas phase with $\mu = 1 \text{ D}$ when the separation is 0.5 nm at 298 K. Compare this energy with the average molar kinetic energy of the molecules.

E14B.5(a) Calculate the average dipole–induced-dipole interaction energy, in joules per mole (J mol^{-1}), between a water molecule and a benzene molecule separated by 1.0 nm.

E14C.1(a) Calculate the vapour pressure of a spherical droplet of water of radius 10 nm at 20 °C. The vapour pressure of bulk water at that temperature is 2.3 kPa and its mass density is 0.9982 g cm^{-3} .

E14C.2(a) The contact angle for water on clean glass is close to zero. Calculate the surface tension of water at 20 °C given that at that temperature water climbs to a height of 4.96 cm in a clean glass capillary tube of internal radius 0.300 mm. The mass density of water at 20 °C is 998.2 kg m^{-3} .

E14C.3(a) Calculate the pressure differential of water across the surface of a spherical droplet of radius 200 nm at 20 °C.

E14C.4(a) The contact angle for water on glass is close to zero. Calculate the surface tension of water at 25 °C given that at that temperature, water climbs to a height of 5.89 cm in a clean glass capillary tube of internal diameter 0.500 mm. The mass density of water at 25 °C is 9970 g cm^{-3} .